

	1	2	3	4	5	6	7	8
A	1. Connector & input protection Field connector, 2-stage bidirectional TVS surge protection, chassis isolation : HMT500(260313A)_connector.kicad_sch		2. Power eFuse (reverse/OV/UV), 60V buck to 5V, LDO 3.3V : HMT500(260313A)_power.kicad_sch		3. MCU STM32G0B1CCT6, SWD, reset, internal temperature sensor : HMT500(260313A)_mcu.kicad_sch		4. Measurement Capacitive humidity sensor (PCAP04) and Pt1000 2-wire (ADS1220) : HMT500(260313A)_measurement.kicad_sch	
B								
C	5. Analog outputs 2x DAC8760 V/I output with TPS26611 miswiring protection : HMT500(260313A)_analog_out.kicad_sch		6. RS-485 THVD2410 +/-70V fault-protected transceiver : HMT500(260313A)_rs485.kicad_sch		7. Aux AO supply VAO 16.1 V (buck), supply monitor : HMT500(260313A)_aux.kicad_sch			
D	DOTECH HMT500(260313) – Oil Moisture Transmitter – Schematic v0.9 – Project No. HMT500(260313A) File names (schematic, PCB, Gerber) = HMT500(260313A).* PCB silkscreen marking = HMT500(260313A) Signal flow: J1 field connector -> protection -> eFuse -> 5 V buck -> 3.3 V LDO; sensor head J3 -> PCAP04 / ADS1220 -> MCU -> DAC8760 x2 / THVD2410 -> J1 Outputs: RS-485 Modbus RTU + 2x analog (4-20 mA / 0-20 mA / 0-10 V / 0-5 V selectable). Supply 12-28 V DC. v0.10: AO D30/D40 TVS1401 +/-14 V. AO +/-28/30 V continuous miswiring is NOT guaranteed (#41). Power symbols: GND, +3V3, +3V3A (analog 3.3 V), +5V, VIN_P (protected input), CHASSIS. Inter-sheet signals: global labels. History v0.3-v0.8 (MCU G0B1, JST harness, DAC8760/TPS26611 datasheet pinouts, RTN isolation, gated DAC SCLK ...): docs/decision-log.md #9-#26. v0.9: 3-pass review (docs/hw/final-review-260313A.md): OVP 32.6 V (R5 36.5k); AO supply VAO 16.1 V buck U15 (DAC AVDD, TPS26611 +Vs; dec. #37, IVS320 AO); D3 series Schottky; RS-485 TVS D60/D61; D2 -> TVS3301; eFuse IN C2; buck CIN: RS485_DE 10k pull-down (in-house bootloader, dec. #31); THVD2410; X7R only; R2 HV 1206; SGOOD -> PA0/PA4; CS_CDC -> PA15; VIN_SENSE PB0; LATCH/CS pull-ups; OPA197 V+ = DAC AVDD; R31/R41 100k; SWD no-connect flags removed; sheet Aux: +Vs clamp, VIN monitor (DAC boost / HSE / sensor ESD: no room, dec. #32). Footprints: KiCad 7.0.11 library + project library HMT500_260313A (GDT Bourns 2035-xx-SM, TVS3301 pad floating). 표기 규칙: 부품번호 = 굵은 글자 (R1, U4) / 부품값 = 보통 글자 (10k, DAC8760) / 전원 네트 = 기울인 글자 (+3V3, VIN_P) 신호 네트 = 테두리 있는 라벨 (SPL_SCK, OUT1_EXT) - 네트 이름은 부품번호·부품명과 겹치지 않게 지음 라벨 모양 = 신호 방향: 뾰족한 쪽이 밖 = 이 시트에서 내보냄(출력), 안 = 받음(입력), 양쪽 = 양방향, 네모 = 아날로그·수동 글자 겹침: 생성기가 부품 몸체·핀·선·라벨·메모와 겹치지 않는 자리에 자동 배치하고, 겹치면 생성 실패로 처리							
E								
F	1	2	3	4	5	6	7	8

Pinouts checked vs manufacturer datasheets 2026-09-30 (STM32: KiCad lib)

Design notes: docs/hw/circuit-design.md

Project No. HMT500(260313A) – HMT500(260313) oil moisture transmitter

DOTECH Co., Ltd.

Sheet: /

File: HMT500(260313A).kicad_sch

Title: HMT500(260313A) Top

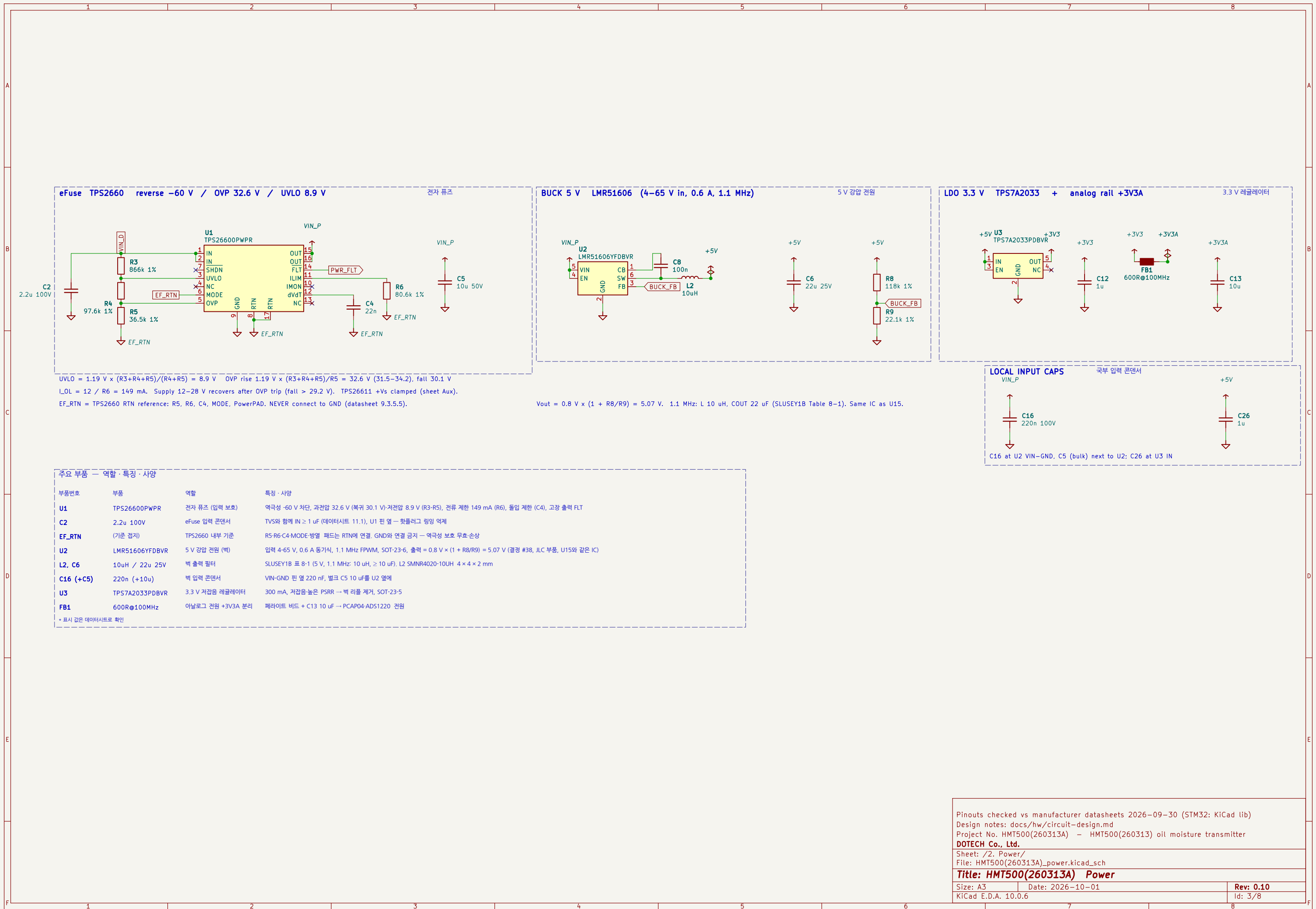
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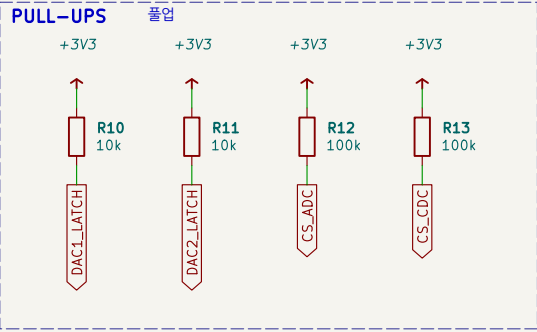
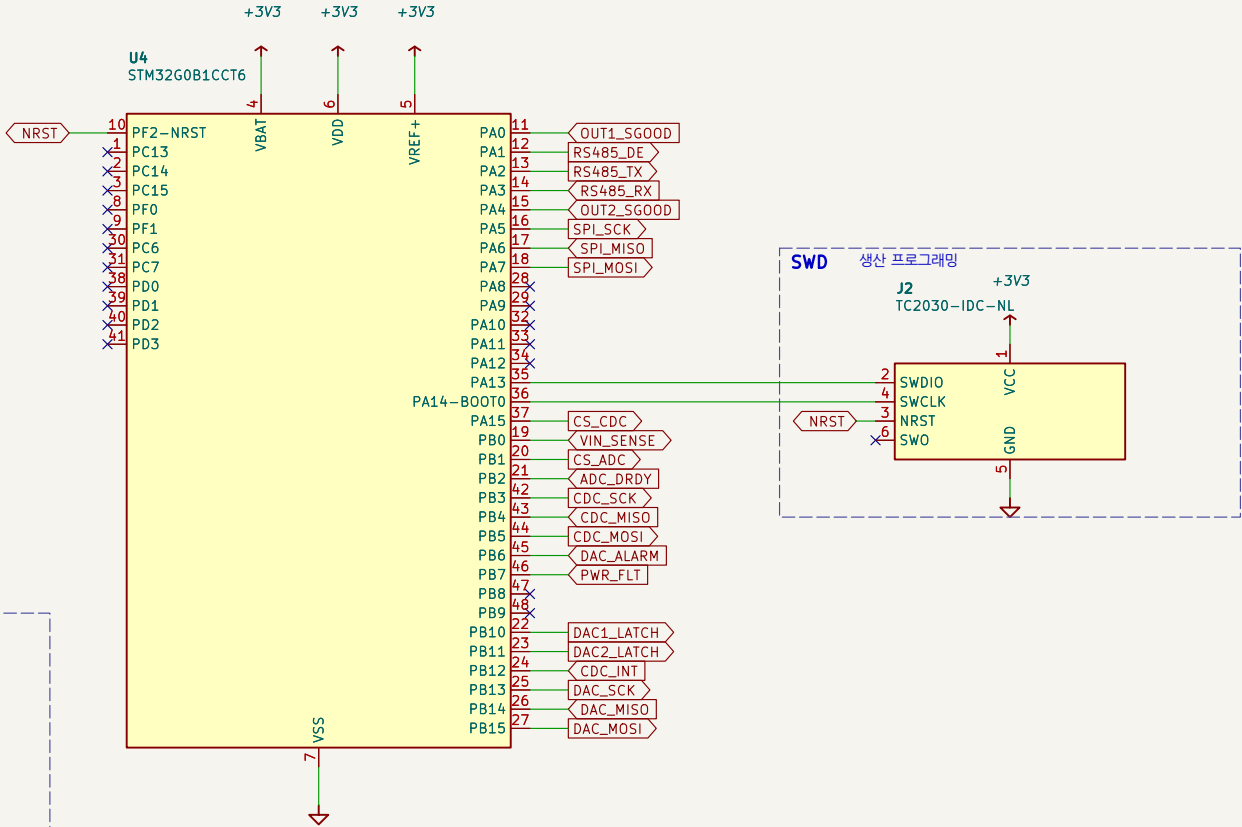
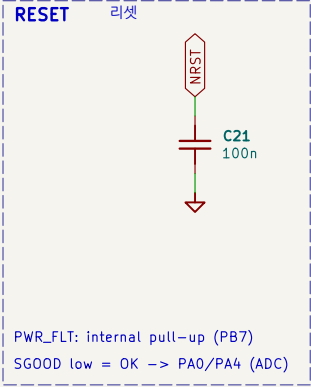
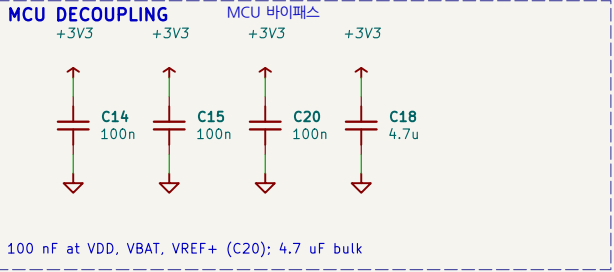
Date: 2026-10-01

Rev: 0.10

KiCad E.D.A. 10.0.6

Id: 1/8





MCU NOTES (STM32G0B1) MCU 설정 메모

BOOT0 shares PA14/SWCLK - no pull-down.

Option bytes: nBOOT_SEL=1, nBOOT0=1, IWDG_SW=0, RDP 1

-> in-house RS-485 bootloader 0x08000000 (dec. #31).

VDDA = VDD (LQFP48); VREF+ tied to +3V3 (C20).

PCB temp (mid): internal sensor, ADC ch TS + TS_CAL.

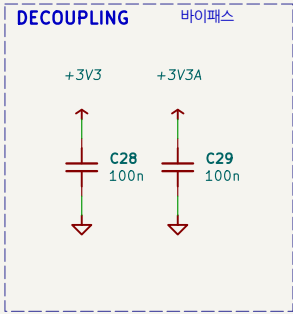
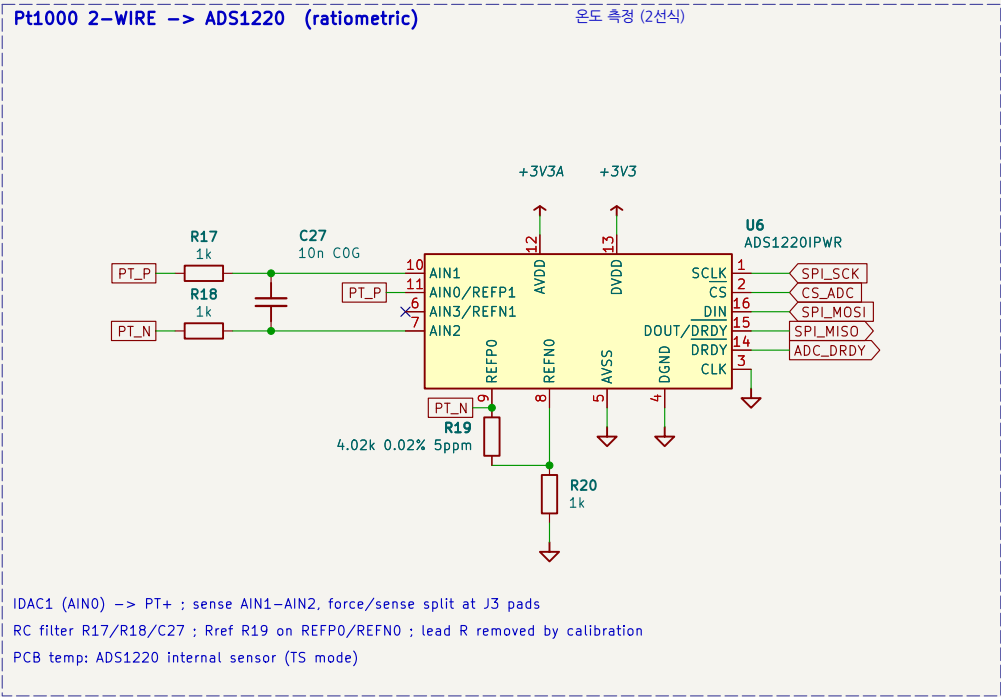
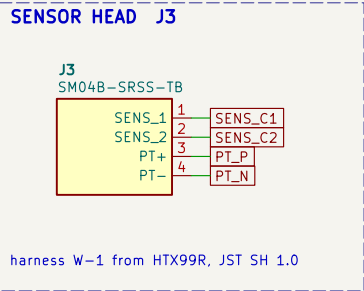
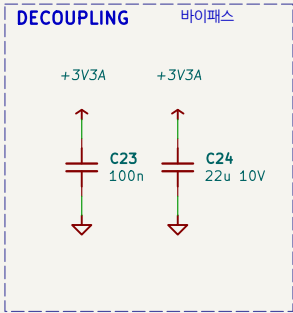
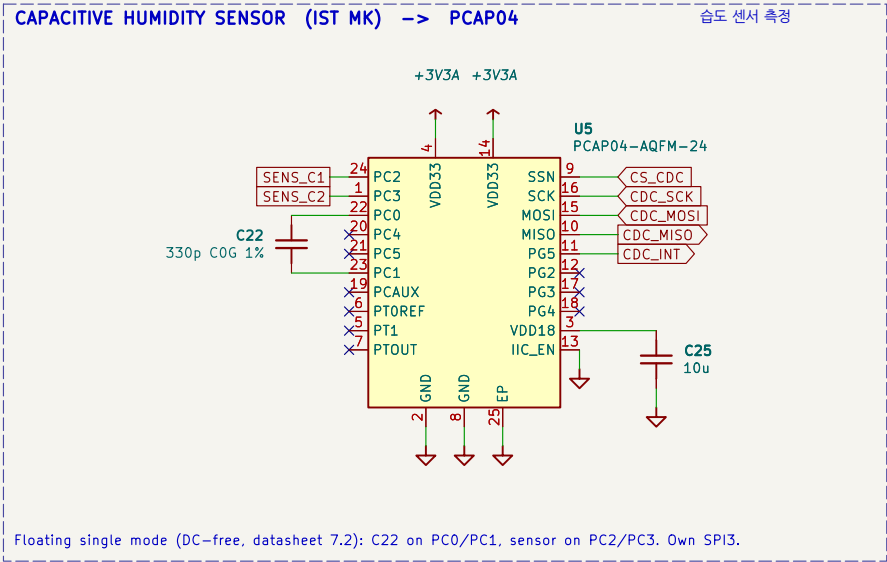
Cortex-M0+: no SWO (J2 pin 6 NC).

Same die as DP2000 (G0B1CCT6); T3 = -40..125 C.

VIN_SENSE (PB0 ADC_IN8) = VIN_P / 11.

주요 부품 — 역할 · 특징 · 사양			
부품번호	부품	역할	특징 · 사양
U4	STM32G0B1CCT6	계어통신·보정 연산	Cortex-M0+ 64 MHz, 플래시 256 KB, RAM 144 KB, -40-85 °C (샘플 #39), LQFP48, 내부 온도센서 (PCB 온도), SPI1 = ADS1220, SPI2 = DAC (모드 3, SCLK 게이트), SPI3 = PCAP04
J2	TC2030-IDC-NL	생산 때 펌웨어 기록 (SWD)	PCB 패드만 (부품 없음). 이후 업데이트는 RS-485 부트로더
C21	100n	리셋 노이즈 필터	NRST 핀, 물딩 후 EMC 여유
R10, R11	10k	DAC LATCH 풀업	리셋 중 LATCH High -> SCLK 게이트 닫힘, DAC 오입력 방지 (v0.9)
R12, R13	100k	CS 풀업	리셋 중 ADS1220-PCAP04 선택 해제 (디지털 입력 부동 금지)

* 표시 값은 데이터시트로 확인



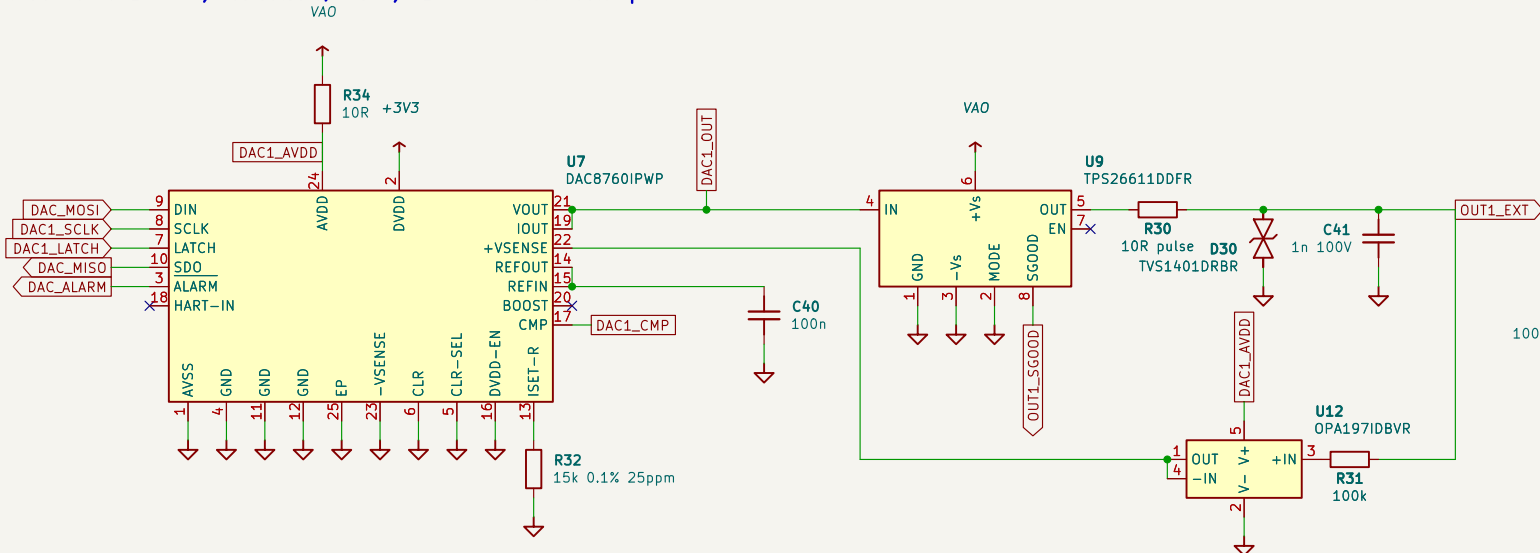
주요 부품 — 역할 · 특징 · 사양			
부품번호	부품	역할	특징 · 사양
J3	SM04B-SRSS-TB	센서 하네스 W-1 연결	JST SH 1.0 mm 4P 옆 삽입, 1-2 = MK33, 3-4 = Pt1000
U5	PCAP04-AQFM-24	습도 센서 정전용량 측정	정전용량-디지털 변환, 기준 C와 비율 측정(방전 시간), DC 없는 구동, 내부 DSP, 전용 SPI3 (모드 1)
C25, C24	10u / 22u	PCAP04 전원 버퍼	VDD18 ≥ 4.7 uF, VDD33 ≥ 10 uF (데이터시트 표 2) — DC 바이어스 후에도 총축 (v0.9)
C22	330p COG 1%	PCAP04 기준 용량	센서 MK33-W (300 pF, 결정 #33) 측정 범위 가운데 값, COG
U6	ADS1220IPWR	Pt1000 온도 측정	24비트 ADC, 50/60 Hz 제거, 내부 온도센서 (PCB 온도), 설정: IDAC = 250 uA 고정 (100 uA면 기준 0.40 V < 권장 0.75 V), PGA ≤ 2 (SP1 모드 1)
R19	4.02k 0.02% 5ppm	Pt1000 비율 측정 기준저항	IDAC 전류 오차 상쇄, 온도계수 5 ppm/°C
R17, R18, C27	1k / 10n COG	입력 RC 필터	차동 노이즈 제거 (COG, 데이터시트 9.2)
* 표시 같은 데이터시트로 확인			

Pinouts checked vs manufacturer datasheets 2026-09-30 (STM32: KiCad lib)
Design notes: docs/hw/circuit-design.md
Project No. HMT500(260313A) — HMT500(260313) oil moisture transmitter
DOTECH Co., Ltd.

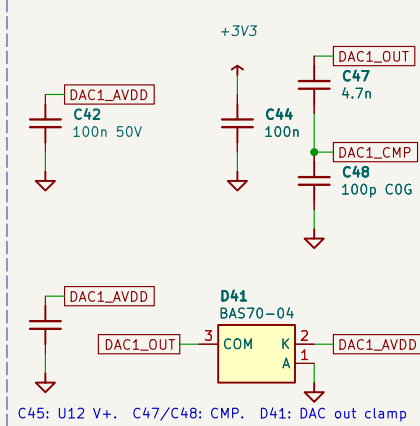
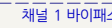
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File: HMT500(260313A)_measurement.kicad_sch

Title: HMT500(260313A) Measurement

Size: A3	Date: 2026-10-01	Rev: 0.10
KiCad E.D.A. 10.0.6		Id: 5/8



VOUT + IOUT one terminal (datasheet 9.1.1.3). +VSENSE = U12 OPA197 follower of terminal OUT1_EXT (after R30) via R31:
no +VSENSE 60k leakage in current mode, series-R drop corrected in voltage mode. ISET-R = R32 15k 0.1%. SG00D low = OK.



C45: U12 V+. C47/C48: CMP. D41: DAC out clamp

DAC SELECTION (decision #10)

DAC 선정

U7/U8 fitted: DAC8760 (16-bit, I + V output).

Drop-in alternates (same pinout, no layout change):

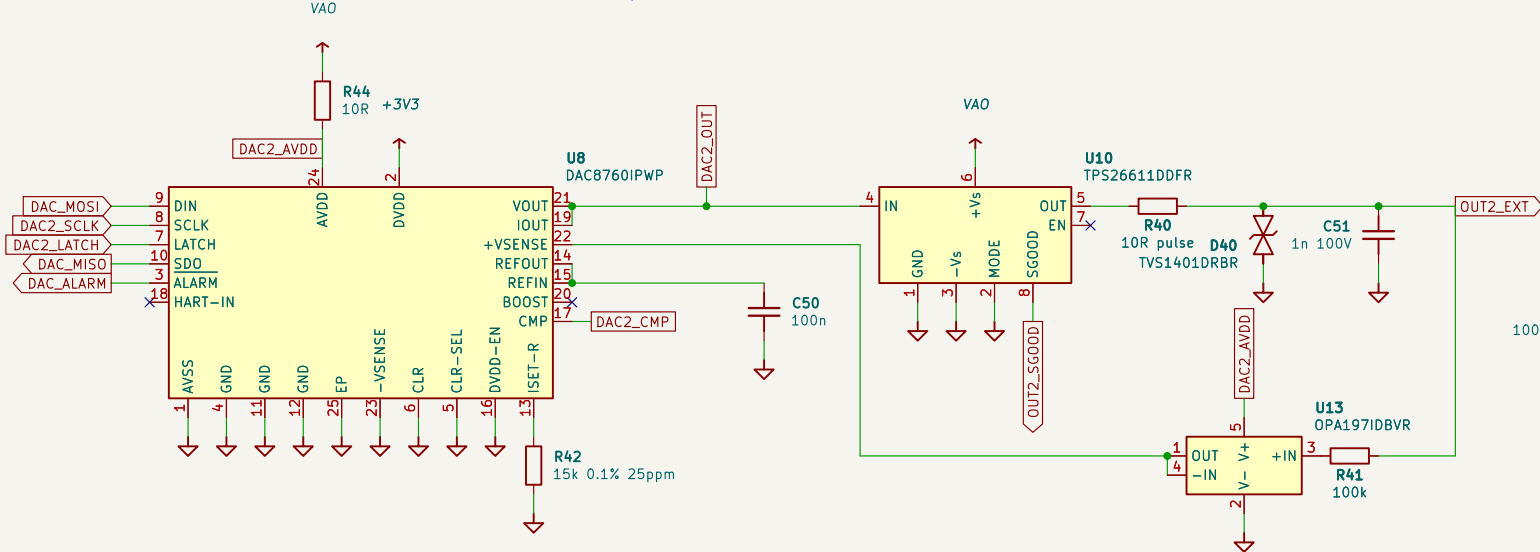
DAC7760 12-bit, I + V (cost-down, V/I kept)

DAC8750 16-bit, I only 4-20/0-20/0-24 mA (EE364-equiv.)

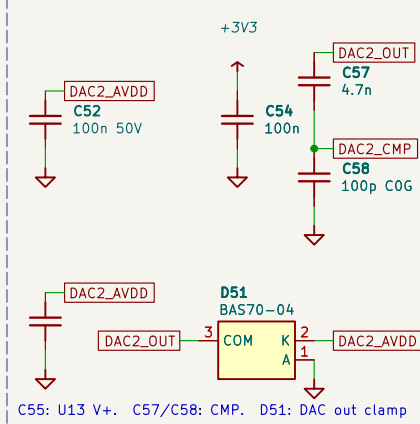
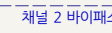
DAC7750 12-bit, I only (-5 uA/step vs EE364 2 uA)

Current-only (x750): VOUT/+VSENSE/-VSENSE absent –

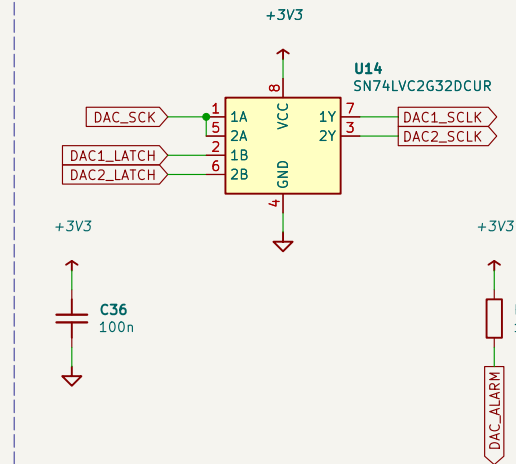
check pin handling before fitting (VERIFY).



VOUT + IOUT one terminal (datasheet 9.1.1.3). +VSENSE = U13 OPA197 follower of terminal OUT2_EXT (after R40) via R41:
no +VSENSE 60k leakage in current mode, series-R drop corrected in voltage mode. ISET-R = R42 15k 0.1%. SG00D low = OK.



C55: U13 V+. C57/C58: CMP. D51: DAC out clamp



which channel -> read status register.

주요 부품 - 역할 · 특징 · 사양

부품번호	부품	역할	특징 · 사양
U7, U8	DAC8760IPWP	아날로그 출력 1~2	16비트, 4-20 / 0-20 / 0-24 mA · 0.5 / 0-10 V 선택, 내부 기준, 고정 알람 — 대체 DAC7760/8750/7750
U9, U10	TPS26611DDFR	출력 오걸선 보호	+50 V, 단자 > +Vs 또는 < -0.2 V면 5 μ s 차단-자동 복귀, 전류 제한 32 mA, RON 7.5 Ω , SGOOD Low = 정상. +Vs = VAO 16.1 V (백, 시트 Aux, 결정 #37)
R30, R40	10R pulse (2512)	서지 전류 제한	출력 보호기와 TVS 사이 직렬
D30, D40	TVS1401DRBR	출력 서지 클램프	± 14 V 동작, 클램프 최대 23.55 V @ 30 A/125 C. AO $\pm 28/30$ V 지속 오걸선 보장 제외 (#41)
C41, C51	1n 100V	출력 고주파 필터	케이블로 들어오는 RF 억제
U12, U13	OPA197IDBVR	+VSENSE 버퍼 (이득 1)	36 V, 입출력 레일투레일 — 단자 전압을 +VSENSE로, 전류 모드 누설 0. V+ = DAC AVDD
R31, R41	100k	버퍼 입력 보호	R30-R40 뒤 단자 전압을 OPA197 + 입력으로 (서지 전류 제한, ± 30 V 오버전 9 mW, v0.9)
R32, R42	15k 0.1% 25ppm	외부 ISET-R (전류 기준)	내부 저항보다 온도 드리프트 작음 (REXT 비트 설정)
R34, R44	10R	DAC AVDD 직렬	VAO → AVDD, 전원 상승 속도 제한 (데이터시트 10 μ s)
C47/C48, C57/C58	4.7n / 100p C0G	DAC CMP 보상	VOUT-CMP 4.7 nF + CMP-GND 100 pF (SBAS528D 8.3.2: 470 pF 초과 시 100 pF 추가) — 케이블 용량 보호기 RON 포함 루프 안정 (IVS320 AO rev 1.0)
D41, D51	BAS70—04	DAC 출력 클램프	DAC 출력 클램프 GND - 0.3 V ~ AVDD + 0.3 V로 — TPS26611 차단-재연결 순간 DAC 핀 보호 (외부 검수 H03, IVS320과 동일)
U14	SN74LVC2G32DCUR	DAC별 SCLK 게이트	OR 2회로: DACn_SCLK = SCK OR LATCHn (SPI2 모드 3)
R33	10k	ALARM 플립	오픈 드레인 ALARM 두 개를 한 선으로 (데이터시트: 외부 10 k Ω)

* 표시 값은 데이터시트로 확인

Pinouts checked vs manufacturer datasheets 2026-09-30 (STM32: KiCad lib)
Design notes: docs/hw/circuit-design.md
Project No. HMT500(260313A) - HMT500(260313) oil moisture transmitter
DOTECH Co., Ltd.

Sheet: /5. Analog outputs/
File: HMT500(260313A)_analog_out.kicad_sch

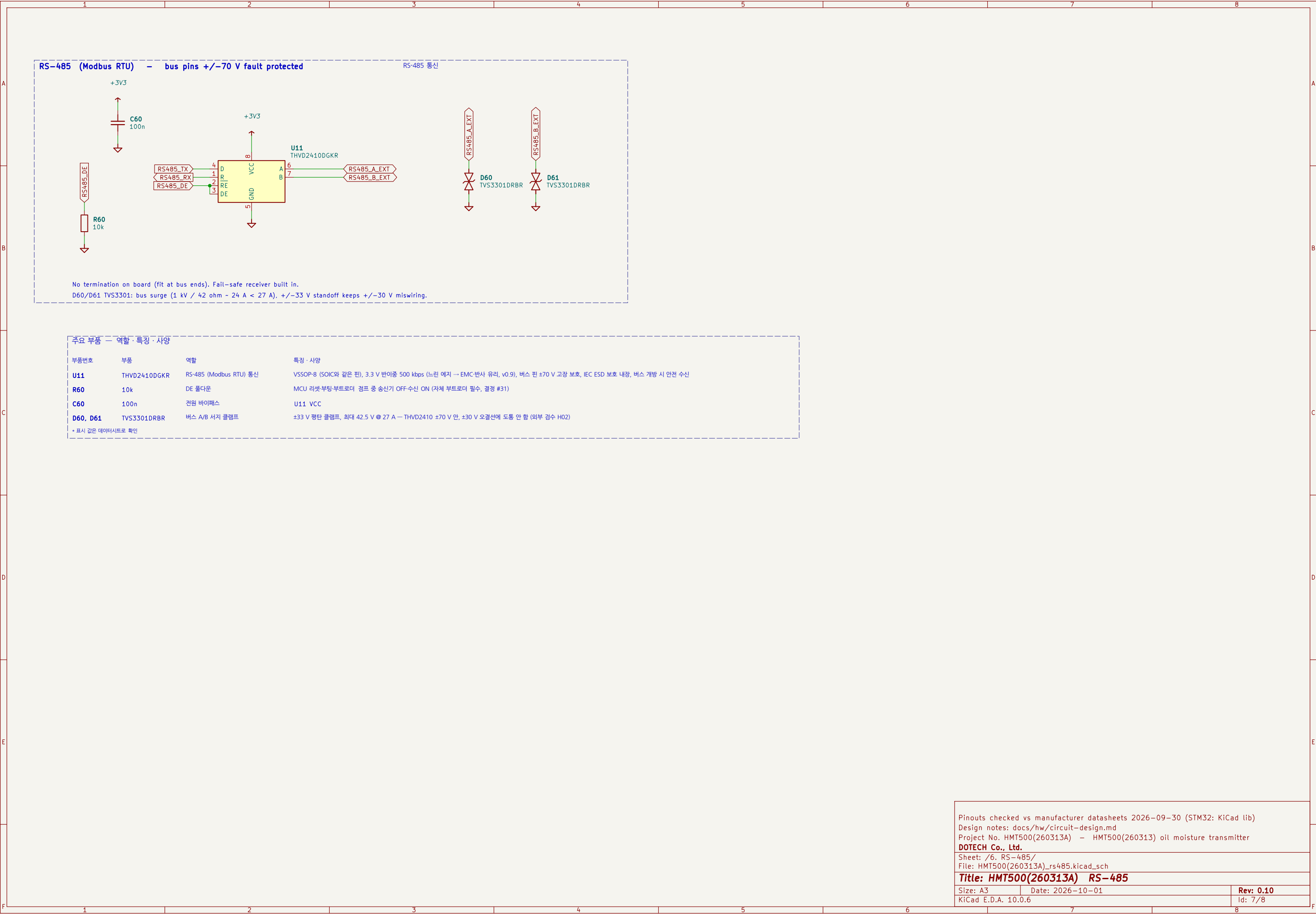
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Size: A3	Date: 2026-10-01
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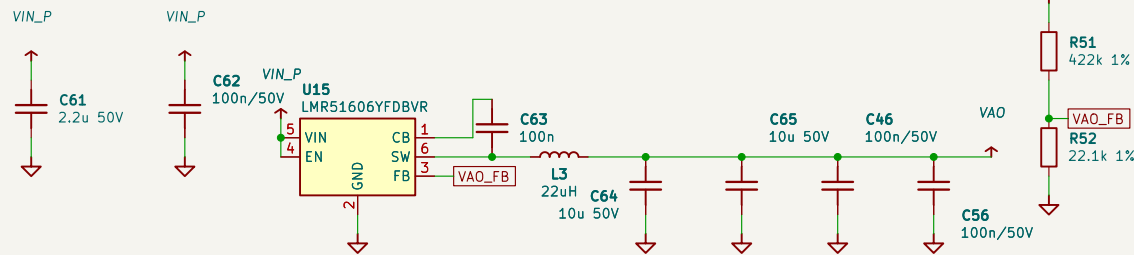
Sheet No.	Date: 2020-10-01
KiCad E.D.A. 10.0.6	

Rev: 0.10

Id: 6/8



AO SUPPLY VAO 16.1 V - DC-DC (decision #37)



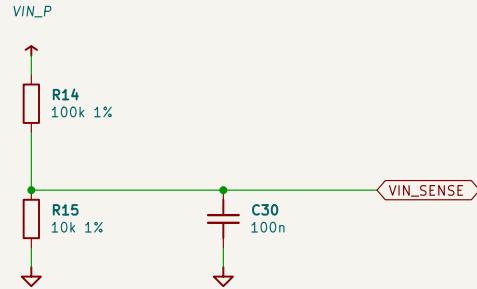
VAO = 0.8 V x (1 + 422k/22.1k) = 16.08 V (15.5–16.6 V). LMR51606 FPWM 1.1 MHz, max duty 98 %.

Load 2 x (DAC 27 + OPA197 1.3 + TPS26611 1.75 mA) = 60 mA. C46/C56 at U9/U10 pin 6.

VIN_P < ~17 V: VAO follows VIN_P (dropout). 24 mA x 500 ohm OK down to VIN ~15.8 V.

SUPPLY MONITOR

전원 전압 감시



VIN_SENSE = VIN_P / 11 -> PB0 (ADC_IN8)

OPTIONS NOT FITTED ON PCB

PCB에 넣지 않은 선택 사항

DAC8760 IOUT boost: not needed – AVDD = VAO 16.1 V (decision #37) cuts DAC heat

from about 0.66 W to 0.39 W per channel (24 mA short, any supply 12–28 V).

Thermal test on a potted prototype still planned.

HSE crystal: no room next to PF0/PF1 -> factory HSI trim + firmware HSI calibration

from Modbus frame timing (USART auto-baud / timer capture).

Sensor-line ESD array: no room next to J3 -> probe ESD test (IEC 61000-4-2) decides;

if needed: protect at the HTX99R side or in the next PCB rev.

주요 부품 — 역할 · 특징 · 사양

부품번호	부품	역할	특징 - 사양
U15	LMR51606YFDBVR	AO 전원 VAO 16.1 V 벅	4-65 V 입력 0.6 A, 1.1 MHz FPMW, SOT-23-6 — DAC AVDD-TPS26611 +Vs-OPA197 V+ 공용 (IVS320 AO rev 1.0과 같은 회로, 결정 #37)
L3, C64, C65	22uH / 10u 50V x2	벅 출력	22 uH 1.05 A 4 × 4 × 2 mm, 10 uF 50 V X7R 1206 × 2 (16 V에서 실효 약 4 uF × 2)
R51, R52	422k / 22.1k 1%	VAO 설정	$0.8 \text{ V} \times (1 + 422/22.1) = 16.08 \text{ V}$
C61, C62, C63	2.2u 50V / 100n / 100n	입력-부트스트랩	VIN 핀 옆 입력 C, CB-SW 100 nF
C46, C56	100n/50V	+Vs 바이패스	C46 → U9 6번, C56 → U10 6번 핀 옆
R14, R15, C30	100k / 10k / 100n	전원 전압 감시	VIN_P / 11 → PB0 ADC (UVLO 8.9 V ~ DAC 최소 10 V 사이 판정)

* 표시 값은 데이터시트로 확인

Pinouts checked vs manufacturer datasheets 2026-09-30 (STM32: KiCad lib)
Design notes: docs/hw/circuit-design.md
Project No. HMT500(260313A) - HMT500(260313) oil moisture transmitter

Sheet: /7. Aux/
File: HMT500(260313A)_aux.kicad_sch

Title: HMT500(260313A) Aux

Size: A3	Date: 2026-10-01	Rev: 0.10
KiCad E.D.A. 10.0.6		Id: 8/8